Supplementary Information: Covariant Field-Theoretic Derivations and Analytical Proofs

Accompanying the manuscripts: "Topological Saturation of Spatial Metrics Unifies Galactic Kinematics and Strong Gravitational Lensing" and "The Topological Saturation Model (ΛTSM)"

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Section S1: Holographic Horizon Derivation of a_0, Virial Relaxation Gating, and Spheroidal Flattening Projection

S1.1 Derivation of the Fundamental Acceleration Threshold (a_0) from Causal Hubble Horizon Closure

Consider a spatially flat ($k = 0$) Friedmann–Lemaître–Robertson–Walker (FLRW) background metric describing late-universe cosmological expansion [1,2]:

$$ds^2 = -c^2dt^2 + a(t)^2\left(dr^2 + r^2 d\Omega^2\right) \tag{S1}$$

where $a(t)$ is the dimensionless cosmic scale factor, r is the comoving radial coordinate, and $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the metric on the unit two-sphere. Defining the physical areal radius relative to a central observer as $R(t,r) \equiv a(t)r$, its exact total differential is $dR = \dot{a}r\,dt + a\,dr = H(t)R\,dt + a\,dr$, where $H(t) \equiv \dot{a}/a$ is the Hubble parameter. Substituting $a\,dr = dR - H(t)R\,dt$ into the FLRW line element transforms the metric into physical Painlevé–Gullstrand-type cosmological coordinates:

$$\begin{aligned} ds^2 = &-\left(1 - \frac{H(t)^2R^2}{c^2}\right)c^2dt^2 - \frac{2H(t)R}{c}c\,dt\,dR \\ &+ dR^2 + R^2 d\Omega^2 \end{aligned} \tag{S2}$$

In the late-universe regime ($H(t) \to H_0$), the causal Hubble horizon (R_H) is defined as the bounding null apparent horizon surface where the timelike metric component vanishes ($g_{tt} = 0$):

$$1 - \frac{H_0^2R_H^2}{c^2} = 0 \implies R_H = \frac{c}{H_0} \tag{S3}$$

The radial gravitational surface gravity (κ_H) evaluated at the causal horizon boundary $R = R_H$ is given by the radial gradient of the temporal metric potential [1,2]:

$$\begin{aligned} \kappa_H &= \frac{c^2}{2}\left|\frac{\partial g_{tt}}{\partial R}\right|_{R=R_H} = \frac{c^2}{2}\left(\frac{2H_0^2R_H}{c^2}\right) \\ &= H_0^2R_H = cH_0 = \frac{c^2}{R_H} \end{aligned} \tag{S4}$$

When this radial horizon tension κ_H undergoes topological closure around a maximal great-circle geodesic loop C_H bounding the causal horizon two-sphere (with physical circumference $L_H = 2\pi R_H$), single-valued periodic phase continuity over a full 2π azimuthal cycle partitions the horizon surface gravity uniformly across the angular manifold:

$$\oint_{C_H} a_0\, d\theta = \kappa_H \implies 2\pi a_0 = cH_0 \tag{S5}$$

Solving for a_0 yields the exact, parameter-free geometric identity governing the fundamental acceleration floor of the spatial metric:

$$a_0 = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H} \tag{S6}$$

Substituting the global late-universe Hubble constant extracted from the unperturbed Hubble-flow regime ($H_0 = 73.0$ km s^{-1} Mpc^{-1} $= 2.3658 \times 10^{-18}$ s^{-1}) [3,4] and the vacuum speed of light ($c = 2.99792458 \times 10^8$ m s^{-1}) yields:

$$\begin{aligned} a_0 &= \frac{(2.99792458 \times 10^8\ \text{m s}^{-1})(2.3658 \times 10^{-18}\ \text{s}^{-1})}{2\pi} \\ &= 1.1288 \times 10^{-10}\ \text{m s}^{-2} \end{aligned} \tag{S7}$$

S1.2 Derivation of the Kinematic Isolation Radius ($D > 20$ Mpc) and the Virial Relaxation Gate (Θ_{relax})

Topological metric saturation is an equilibrium boundary phenomenon that requires the local spatial manifold to be free of net kinematic collapse or violent multi-node collisions. Let the cosmological Hubble recession velocity at distance D be $v_H(D) = H_0D$, and let v_{pec} denote the local peculiar infall velocity. The local volume expansion scalar of the manifold fluid four-velocity u^μ, defined as $\theta_{\text{expansion}} \equiv \nabla_\mu u^\mu = \frac{3H_0}{c} + \nabla\cdot\left(\frac{\vec{v}_{\text{pec}}}{c}\right)$, maintains positive net metric dilation ($\theta_{\text{expansion}} > 0$) if and only if:

$$v_H(D) > |v_{\text{pec}}| \implies H_0D > |v_{\text{pec}}| \tag{S8}$$

Given the characteristic maximum peculiar velocity dispersion observed in rich local supercluster infall regions ($\sigma_{\text{pec,max}} \approx 1500$ km s^{-1}), the critical kinematic isolation distance beyond which the global Hubble flow strictly dominates local peculiar noise is:

$$D_{\text{crit}} = \frac{\sigma_{\text{pec,max}}}{H_0} = \frac{1500\ \text{km s}^{-1}}{73.0\ \text{km s}^{-1}\ \text{Mpc}^{-1}} \approx 20.5\ \text{Mpc} \tag{S9}$$

Conversely, on the intergalactic scale of the Local Group binary system (Milky Way–Andromeda, separated by $D \approx 0.77$ Mpc) [5], the background cosmological recession is $v_H = 73.0 \times 0.77 \approx 56.2$ km s^{-1}, whereas the mutual baryonic infall velocity is $|v_{\text{pec}}| \approx 110$ km s^{-1} $> v_H$. Because the intergalactic bridge undergoes net kinematic contraction ($\theta_{\text{expansion}} < 0$) and comprises $N_{\text{nodes}} = 2$ unmerged mass centers violating single-node virial equilibrium ($2K + U = 0$), the macroscopic thermodynamic gating functional vanishes identically across the intergalactic bridge:

$$\Theta_{\text{relax}} \equiv \Theta(v_H - |v_{\text{pec}}|)\cdot\delta_{N_{\text{nodes}},1} = 0 \tag{S10}$$

where $\Theta(\cdot)$ denotes the Heaviside step function and $\delta_{i,j}$ is the Kronecker delta. In continuous differential field theory, this macroscopic gate is the sharp-boundary limit ($\theta_{\text{tol}} \to 0^+, \sigma_{\text{vir}} \to 0^+$) of the smooth C^∞ covariant relaxation functional $\Theta^{\text{cov}}_{\text{relax}}(x^\mu) = \frac{1}{2}\left[1+\tanh(\theta_{\text{expansion}}/\theta_{\text{tol}})\right]\exp(-\mathcal{V}^2_{\text{vir}}/\sigma^2_{\text{vir}})$, where $\mathcal{V}_{\text{vir}} \equiv |2K+U|/|U|$ measures the local departure from virial equilibrium. Consequently, while each individual virialized galaxy ($\mathcal{V}_{\text{vir}} \to 0, N_{\text{nodes}} = 1$ internally) maintains an active internal saturation envelope, the contracting space between them ($\theta_{\text{expansion}} < 0$) and the outer Milky Way boundary interacting with the Magellanic Clouds [6] experience 0% topological saturation and obey standard Newtonian attraction.

S1.3 Analytical Derivation of the Flattening Projection Factor ($\sqrt{1-\eta^2}$) via Oblate Maclaurin Spheroid Potential Theory

Consider a self-gravitating baryonic system distributed within an axisymmetric oblate Maclaurin spheroid with semi-major equatorial axes $a_x = a_y = a$ and semi-minor vertical polar axis $a_z = b$ ($b \le a$), characterized by the dimensionless morphological aspect ratio:

$$\eta \equiv \frac{b}{a} \in [0,1] \tag{S11}$$

In Cartesian coordinates, the bounding surface of the spheroidal manifold is given by:

$$\frac{x^2+y^2}{a^2} + \frac{z^2}{b^2} = 1 \tag{S12}$$

In classical potential theory, the gravitational and geometric anisotropy of an oblate spheroid is governed entirely by the first meridional eccentricity (e) of its vertical cross-section:

$$e \equiv \sqrt{1-\left(\frac{b}{a}\right)^2} = \sqrt{1-\eta^2} \tag{S13}$$

By Maclaurin's theorem, the internal gravitational potential $\Phi(r,z)$ of an oblate spheroid of density ρ_0 decomposes into radial and vertical harmonic coefficients $A_1(e)$ and $A_3(e)$ that depend solely on e:

$$\begin{aligned} A_1(e) &= \frac{\sqrt{1-e^2}}{e^3}\arcsin(e) - \frac{1-e^2}{e^2}, \\ A_3(e) &= \frac{2}{e^2} - \frac{2\sqrt{1-e^2}}{e^3}\arcsin(e) \end{aligned} \tag{S14}$$

satisfying the trace identity $2A_1(e) + A_3(e) = 2$. When the isotropic background horizon tension a_0 couples to the internal dynamics of the spheroid, the active planar strain tensor projected onto the equatorial plane ($z = 0$) is proportional to the geometric eccentricity operator $e = \sqrt{1-\eta^2}$ measuring the departure from three-dimensional spherical isotropy:

$$a_{0,\text{eff}}(\eta) = a_0\, e = a_0\sqrt{1-\eta^2} \tag{S15}$$

This exact geometric projection analytically governs the two observational limiting regimes:

1. **Quasi-2D Thin Disk Regime ($\eta = b/a \to 0$):** The meridional eccentricity approaches unity ($e = \sqrt{1-0} = 1$), yielding $a_{0,\text{eff}} \to a_0$ and triggering 100% internal planar saturation.
2. **Diffuse 3D Spherical Regime ($\eta = b/a \to 1$, e.g., ultra-diffuse galaxy NGC 1052-DF2 [7] or globular clusters):** The meridional eccentricity vanishes identically ($e = \sqrt{1-1} = 0$), yielding:

$$\lim_{\eta \to 1} a_{0,\text{eff}}(\eta) = 0 \tag{S16}$$

Because the effective internal threshold vanishes ($a_{0,\text{eff}} = 0$), any finite internal baryonic acceleration satisfies $a_{\text{baryonic}} \gg a_{0,\text{eff}}$, rigorously enforcing pure Newtonian kinematics without invoking ad-hoc external field effects.

Section S2: Covariant DBI Action, Stress-Energy Tensor Derivation, Modified Einstein Field Equations, and Tensor Mode Luminosity ($c_T = c$)

S2.1 Total Covariant System Action

We formulate the four-dimensional spacetime manifold with metric tensor $g_{\mu\nu}$ (signature $-,+,+,+$) via the total covariant action S, comprising the Einstein–Hilbert gravitational action, the Dirac–Born–Infeld (DBI) topological phase-transition scalar field ϕ [8–10], and the standard baryonic matter action $\mathcal{L}_{\text{baryon}}(\psi, g_{\mu\nu})$:

$$\begin{aligned} S = \int d^4x\sqrt{-g}\Big[&\frac{c^4 R}{16\pi G} + \mathcal{L}_{\text{DBI}}(\phi, X) \\ &+ \mathcal{L}_{\text{baryon}}(\psi, g_{\mu\nu})\Big] \end{aligned} \tag{S17}$$

where $R = g^{\mu\nu}R_{\mu\nu}$ is the Ricci curvature scalar, $g \equiv \det(g_{\mu\nu})$, and the DBI Lagrangian density is defined as:

$$\mathcal{L}_{\text{DBI}}(\phi, X) = -M^4\sqrt{1-\frac{2|X|}{M^4}} + M^4 - V(\phi) \tag{S18}$$

Here, M represents the fundamental manifold tension scale, $V(\phi)$ is the self-interaction potential (with $+M^4$ explicitly normalizing the zero-gradient vacuum baseline), and $|X|$ denotes the invariant norm ($|X| = X > 0$ along timelike cosmological evolution and $|X| = -X > 0$ along stationary spatial gradients) of the canonical covariant kinetic scalar:

$$X \equiv -\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\,\nabla_\nu\phi \tag{S19}$$

To characterize the non-linear saturation of the spatial gradient, we define the dimensionless relativistic Lorentz saturation factor γ:

$$\gamma \equiv \left(1-\frac{2|X|}{M^4}\right)^{-1/2} \ge 1 \tag{S20}$$

which enforces an absolute upper speed/gradient limit $2|X| \le M^4$ on local manifold deformation.

S2.2 Explicit Variational Derivation of the Phantom Stress-Energy Tensor ($T^{(\phi)}_{\mu\nu}$) and Modified Einstein Equations

Requiring stationarity of the total action under arbitrary variations of the inverse metric tensor $\delta g^{\mu\nu}$ ($\delta S/\delta g^{\mu\nu} = 0$) yields:

$$\delta S_{\text{EH}} + \delta S_{\text{DBI}} + \delta S_{\text{baryon}} = 0 \tag{S21}$$

Evaluating the variation of the DBI action $S_{\rm DBI} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\rm DBI}(\phi,X)$ explicitly:

$$\delta S_{\rm DBI} = \int d^4x \left[(\delta\sqrt{-g})\mathcal{L}_{\rm DBI} + \sqrt{-g}\frac{\partial \mathcal{L}_{\rm DBI}}{\partial X}\delta X \right] \tag{S22}$$

We employ the exact differential tensor identities:

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}, \qquad \delta X = -\frac{1}{2}\nabla_\mu\phi\,\nabla_\nu\phi\,\delta g^{\mu\nu} \tag{S23}$$

Differentiating $\mathcal{L}_{\rm DBI}(\phi,X)$ with respect to the kinetic invariant norm $|X|$ yields:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\rm DBI}}{\partial |X|} &= -M^4 \cdot \frac{1}{2}\left(1-\frac{2|X|}{M^4}\right)^{-1/2}\left(-\frac{2}{M^4}\right) \\ &= \left(1-\frac{2|X|}{M^4}\right)^{-1/2} = \gamma\end{aligned} \tag{S24}$$

Substituting these expressions into the covariant stress-energy tensor definition $T^{(\phi)}_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta S_{\rm DBI}}{\delta g^{\mu\nu}}$ gives:

$$\begin{aligned}T^{(\phi)}_{\mu\nu} &= \gamma\nabla_\mu\phi\,\nabla_\nu\phi + g_{\mu\nu}\,\mathcal{L}_{\rm DBI} \\ &= \gamma\nabla_\mu\phi\,\nabla_\nu\phi + g_{\mu\nu}\left[M^4(1-\gamma^{-1}) - V(\phi)\right]\end{aligned} \tag{S25}$$

Combining this with the Einstein–Hilbert variation $\frac{\delta S_{\rm EH}}{\delta g^{\mu\nu}} = \frac{c^4}{16\pi G}\sqrt{-g}\,G_{\mu\nu}$ and the ordinary baryonic stress-energy tensor $T^{\rm baryon}_{\mu\nu}$ produces the complete Modified Einstein Field Equations:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}R\,g_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\rm baryon}_{\mu\nu} + T^{(\phi)}_{\mu\nu}\right) \tag{S26}$$

This demonstrates rigorously that the apparent "missing mass" in galactic halos and cluster lenses arises purely from the non-linear geometric stress $T^{(\phi)}_{\mu\nu}$ as γ diverges toward the saturation boundary $2|X| \to M^4$, with 0% non-baryonic dark matter particles.

S2.3 Covariant Euler–Lagrange Equation of Motion for ϕ

Varying the total action with respect to the scalar field perturbation $\delta\phi$ ($\delta S/\delta\phi = 0$) yields the curved-spacetime Euler–Lagrange equation:

$$\frac{1}{\sqrt{-g}}\partial_\mu\left[\sqrt{-g}\frac{\partial \mathcal{L}_{\rm DBI}}{\partial(\partial_\mu\phi)}\right] - \frac{\partial \mathcal{L}_{\rm DBI}}{\partial\phi} = J_{\rm baryon} \tag{S27}$$

Applying the chain rule $\frac{\partial \mathcal{L}_{\rm DBI}}{\partial(\partial_\mu\phi)} = \frac{\partial \mathcal{L}_{\rm DBI}}{\partial X}\frac{\partial X}{\partial(\partial_\mu\phi)} = -\gamma\nabla^\mu\phi$, noting that $\frac{\partial \mathcal{L}_{\rm DBI}}{\partial\phi} = -\frac{dV(\phi)}{d\phi} \equiv -V_{,\phi}$, and invoking the covariant divergence identity $\nabla_\mu A^\mu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}A^\mu)$ yields the exact non-linear field equation:

$$\nabla_\mu(\gamma\nabla^\mu\phi) - V_{,\phi} = -J_{\rm baryon} \tag{S28}$$

where $J_{\rm baryon} \equiv -\frac{\delta \mathcal{L}_{\rm baryon}}{\delta\phi} \propto T^\mu_{\mu,\rm baryon} = -\rho_{\rm baryon}c^2$ represents the trace-coupled baryonic source density in the non-relativistic limit.

S2.4 Formal Proof of Luminal Gravitational Wave Propagation ($c_T = c$)

Within the generalized Horndeski scalar-tensor framework (the most general four-dimensional Lorentz-invariant scalar-tensor theory with second-order equations of motion), our action maps onto the structural functions:

$$\begin{aligned}G_2(\phi,X) &= \mathcal{L}_{\rm DBI}(\phi,X), \qquad G_3 = 0, \\ G_4 &= \frac{c^4}{16\pi G} = \text{const}, \quad G_5 = 0\end{aligned} \tag{S29}$$

The propagation speed squared c_T^2 of transverse-traceless tensor perturbations (h_{ij}, gravitational waves) over a cosmological background is governed by the ratio of the tensor kinetic and gradient coefficients:

$$c_T^2 = c^2\frac{\mathcal{F}_T}{\mathcal{G}_T} = c^2\frac{G_4 - X(\ddot{\phi}G_{5,X} + G_{5,\phi})}{G_4 - 2XG_{4,X} + X(H\dot{\phi}G_{5,X} - G_{5,\phi})} \tag{S30}$$

Because $\mathcal{L}_{\rm DBI}$ is minimally coupled to the Ricci scalar R without derivative couplings to the Einstein tensor, we have identically across all phase space $G_{4,X} \equiv 0$ and $G_5(\phi,X) \equiv 0$. Substituting these exact identities into Eq. (S30) yields:

$$c_T^2 = c^2\frac{G_4 - 0}{G_4 - 0 + 0} = c^2 \implies c_T = c \tag{S31}$$

Thus, gravitational waves propagate at the exact vacuum speed of light across all regimes, strictly satisfying the empirical multi-messenger bound from the binary neutron star merger GW170817/GRB 170817A ($|c_T/c - 1| \le 10^{-15}$) [11].

Section S3: Newtonian Gauge Perturbation Expansion, KAM Derivation of φ, Derivation of ν, and Asymptotic Limits

S3.1 Second-Order Taylor Expansion of the DBI Lagrangian in the Conformal Newtonian Gauge

Consider a perturbed spatially flat FLRW metric in the conformal Newtonian gauge (with conformal time $d\tau = dt/a$) [12]:

$$ds^2 = a(\tau)^2\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right] \tag{S32}$$

where Ψ and Φ are the Bardeen gravitational potentials. To first order in the metric potentials, the inverse metric components are $g^{00} = -a(\tau)^{-2}(1-2\Psi)$, $g^{ij} = a(\tau)^{-2}(1+2\Phi)\delta^{ij}$, and $g^{0i} = 0$. Decomposing the DBI scalar field into a homogeneous time-dependent background and a spatiotemporal perturbation, $\phi(\tau,\vec{x}) = \bar{\phi}(\tau) + \delta\phi(\tau,\vec{x})$, we evaluate the kinetic invariant $X = -\frac{1}{2}[g^{00}(\partial_\tau\phi)^2 + g^{ij}\partial_i\phi\,\partial_j\phi]$:

$$X = \frac{1}{2a^2}(1-2\Psi)(\bar{\phi}' + \delta\phi')^2 - \frac{1}{2a^2}(1+2\Phi)(\nabla\delta\phi)^2 \tag{S33}$$

where primes denote derivatives with respect to τ. Expanding and grouping terms by perturbation order yields $X = \bar{X} + \delta X_1 + \delta X_2$, with:

$$\begin{aligned}\bar{X} &= \frac{(\bar{\phi}')^2}{2a^2}, \qquad \delta X_1 = \frac{\bar{\phi}'\delta\phi' - \Psi(\bar{\phi}')^2}{a^2}, \\ \delta X_2 &= \frac{(\delta\phi')^2 - 2\Psi\bar{\phi}'\delta\phi' - (\nabla\delta\phi)^2}{2a^2}\end{aligned} \tag{S34}$$

Performing a Taylor series expansion of $\mathcal{L}_{\rm DBI}(\phi, X)$ around the background state $(\bar{\phi}, \bar{X})$ up to second order using $\frac{\partial \mathcal{L}}{\partial X}\Big|_{\bar{X}} = \bar{\gamma}$ and $\frac{\partial^2 \mathcal{L}}{\partial X^2}\Big|_{\bar{X}} = \frac{\partial \bar{\gamma}}{\partial \bar{X}} = \frac{\bar{\gamma}^3}{M^4}$, we obtain the complete perturbed Lagrangian:

$$\mathcal{L}_{\rm DBI} \approx \left[-M^4\bar{\gamma}^{-1} + M^4 - V(\bar{\phi})\right] + \left[\bar{\gamma}\delta X_1 - V_{,\phi}\delta\phi\right] + \left[\bar{\gamma}\delta X_2 + \frac{\bar{\gamma}^3}{2M^4}(\delta X_1)^2 - \frac{1}{2}V_{,\phi\phi}(\delta\phi)^2\right] \tag{S35}$$

From the quadratic perturbation action, the effective adiabatic sound speed squared of the spatial metric perturbation is $c_s^2 = \frac{\partial P/\partial X}{\partial \rho/\partial X} = \bar{\gamma}^{-2} = 1 - \frac{2\bar{X}}{M^4}$ [10, 13]. As the system approaches topological saturation ($\bar{\gamma} \to \infty$), $c_s^2 \to 0$, arresting dispersive wave propagation and locking the geometric deformation gradient locally around the baryonic source.

S3.2 Mathematical Derivation of the Topological Scaling Constant (φ) via KAM Theory and Diophantine Orbital Stability

In a flattened quasi-2D spiral galaxy ($\eta \to 0$), baryonic surface density self-organizes along an equiangular logarithmic spiral manifold of constant pitch angle ψ [14]: $r(\theta) = r_0 \exp(\theta \cot\psi)$. In action-angle variables $(J_r, J_\theta, \theta_r, \theta_\theta)$, the Hamiltonian of a test particle orbiting within the perturbed disk potential takes the form $\mathcal{H}(J_r, J_\theta, \theta_r, \theta_\theta) = \mathcal{H}_0(J_r, J_\theta) + \varepsilon \sum_{p,q} \mathcal{H}_{p,q}(J_r, J_\theta) \exp[i(p\theta_r - q\theta_\theta)]$, characterized by the epicyclic radial frequency $\Omega_r = \partial\mathcal{H}_0/\partial J_r$ and azimuthal orbital frequency $\Omega_\theta = \partial\mathcal{H}_0/\partial J_\theta$.

In canonical perturbation theory, the generating function contains resonant denominators of the form $(p\Omega_r - q\Omega_\theta)$ for integers $p, q \in \mathbb{Z}$. By the Kolmogorov–Arnold–Moser (KAM) theorem, invariant tori survive destructive Lindblad resonances and guarantee collision-free mass packing at the disk periphery if and only if the dimensionless winding frequency ratio $\omega \equiv \Omega_r/\Omega_\theta$ is sufficiently irrational to satisfy the Diophantine stability condition:

$$\left|\omega - \frac{p}{q}\right| \geq \frac{C(\omega)}{q^2} \qquad \forall\, p, q \in \mathbb{Z},\ q > 0 \tag{S36}$$

By Hurwitz's theorem in Diophantine approximation theory, an irrational number ω expressed as a regular infinite continued fraction $[a_0; a_1, a_2, a_3, \ldots]$ maximizes the Diophantine stability constant ($C_{\max} = 1/\sqrt{5}$, maximizing the metric distance from all destructive rational resonances p/q) if and only if all partial quotients assume their minimal positive integer value $a_k = 1$:

$$\varphi = [1; 1, 1, 1, \ldots] = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}} \implies \varphi^2 - \varphi - 1 = 0 \tag{S37}$$

whose unique positive root is the fundamental topological scaling constant (the Golden Ratio):

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339887\ldots \tag{S38}$$

Consequently, in a relaxed quasi-2D spiral disk ($\eta \to 0$) self-organized into its state of minimal resonant action, the fundamental horizon threshold $a_{0,\rm eff}(\eta \to 0) = a_0$ scales by the optimal Diophantine packing factor φ, defining the planar acceleration threshold:

$$a_\varphi = a_0\varphi = \frac{cH_0\varphi}{2\pi} \approx 1.8264 \times 10^{-10}\ \text{m s}^{-2} \tag{S39}$$

S3.3 Derivation of the Topological Metric Deformation Factor (ν) via Neumann Series and Direct Variational Closure

In the quasi-static planar limit ($z = 0$), the modified gravitational field equation couples the classical Newtonian baryonic acceleration $a_{\rm baryonic}(r) = -\nabla_r \Phi_{\rm baryon}$ to the non-linear geometric polarization of the DBI manifold. Let $\rho(r) \in (0, 1)$ denote the local dimensionless topological susceptibility of the saturated manifold. Because the induced metric strain acts as a self-consistent geometric feedback source within the saturated 2D disk plane, the total effective radial acceleration $a_{\rm eff}(r)$ obeys the linear Fredholm-type self-interaction relation:

$$a_{\rm eff}(r) = a_{\rm baryonic}(r) + \rho(r)\, a_{\rm eff}(r) \tag{S40}$$

Solving for $a_{\rm eff}(r)$ via an exact convergent Neumann geometric series (summing all n-th order recursive metric feedback loops):

$$a_{\rm eff}(r) = a_{\rm baryonic}(r) \sum_{n=0}^{\infty} [\rho(r)]^n = \frac{a_{\rm baryonic}(r)}{1 - \rho(r)} \equiv a_{\rm baryonic}(r)\, \nu(r) \tag{S41}$$

Because gravitational field energy density scales quadratically with field strength ($\mathcal{E}_g \propto g^2 \propto a_{\rm baryonic}$), the canonical field amplitude scales with the square root of acceleration. Defining the normalized dimensionless field amplitude across the planar threshold a_φ as $u(r) \equiv (a_{\rm baryonic}(r)/a_\varphi)^{1/2}$, the spatial attenuation of the saturation susceptibility $\rho(u)$ across energy states follows the canonical first-order differential relaxation law $d\rho/du = -\rho(u)$ subject to $\lim_{u\to 0} \rho(u) = 1$, yielding the exact Boltzmann-type spatial saturation decay factor $\rho(r) = \exp(-u(r)) = \exp[-(a_{\rm baryonic}(r)/a_\varphi)^{1/2}]$. Substituting $\rho(r)$ into the Neumann series resolvent $(1 - \rho)^{-1}$ yields the exact Covariant Topological Metric Deformation Factor (ν):

$$\nu(r) = \frac{1}{1 - \exp\left(-\sqrt{\frac{a_{\rm baryonic}(r)}{a_\varphi}}\right)} \tag{S42}$$

Direct Variational Euler–Lagrange Closure: Crucially, Eq. (S42) also arises directly as the exact first integral of the covariant Euler–Lagrange equation (S28) without invoking a perturbative Neumann expansion. In a stationary galactic potential where the scalar gradient is superimposed on the cosmological background $\bar{\phi}(t)$ (or formulated in terms of the positive-definite spatial gradient invariant $X_s \equiv \frac{1}{2} g^{ij} \partial_i\delta\phi\, \partial_j\delta\phi = \frac{1}{2} a_{\rm eff}^2 > 0$ within the Horndeski $G_2(X_s)$ subclass where $c_T = c$ identically), the static Euler–Lagrange equation (S28) takes the generalized Gauss–Poisson divergence form $\nabla \cdot [\mathcal{L}_{X_s} \nabla\delta\phi] = 4\pi G \rho_{\rm baryon} = -\nabla \cdot \vec{a}_{\rm baryonic}$. Integrating over a Gauss surface of radius r with the effective gravitational dielectric susceptibility $\mathcal{L}_{X_s} = 1 - \rho(r) = 1 - \exp[-(a_{\rm baryonic}(r)/a_\varphi)^{1/2}] = \nu^{-1}(r)$ yields $\nu^{-1}(r)\vec{a}_{\rm eff}(r) = \vec{a}_{\rm baryonic}(r)$, recovering $a_{\rm eff}(r) = \nu(r) a_{\rm baryonic}(r)$ identically from the variational principle.

For circular orbits in the galactic midplane, equating the centripetal acceleration $a_{\rm eff}(r) = \frac{V^2_{\rm calculated}(r)}{r}$ to $a_{\rm baryonic}(r)\nu(r) = \frac{V^2_{\rm baryonic}(r)}{r}\nu(r)$ and multiplying through by r yields the parameter-free rotation velocity law:

$$V_{\rm calculated}(r) = \sqrt{V^2_{\rm baryonic}(r)\,\nu(r)} = \frac{V_{\rm baryonic}(r)}{\sqrt{1-\exp\left(-\sqrt{\frac{a_{\rm baryonic}(r)}{a_\varphi}}\right)}} \tag{S43}$$

S3.4 Analytical Proofs of the Two Asymptotic Regimes: Baryonic Tully–Fisher Relation and Solar System PPN Bounds

(a) Deep Low-Acceleration Limit ($r \to \infty, a_{\rm baryonic} \ll a_\varphi$) — Exact Derivation of the Baryonic Tully–Fisher Relation: At large galactocentric radii beyond the main stellar disk, the Newtonian baryonic acceleration approaches $a_{\rm baryonic}(r) = \frac{GM_{\rm baryon}}{r^2} \ll a_\varphi$. Defining $x \equiv (a_{\rm baryonic}(r)/a_\varphi)^{1/2} \ll 1$ and expanding $1-\exp(-x) = x\left(1-\frac{x}{2}+\mathcal{O}(x^2)\right)$ gives $\nu(r) = \sqrt{\frac{a_\varphi}{a_{\rm baryonic}(r)}} + \frac{1}{2} + \mathcal{O}(x)$. Substituting the leading-order term together with $a_{\rm baryonic}(r) = \frac{GM_{\rm baryon}}{r^2}$ and $V^2_{\rm baryonic}(r) = \frac{GM_{\rm baryon}}{r}$ into Eq. (S43):

$$V_\infty \equiv \lim_{r\to\infty} V_{\rm calculated}(r) = \lim_{r\to\infty}\sqrt{\frac{GM_{\rm baryon}}{r}\cdot\frac{r\sqrt{a_\varphi}}{\sqrt{GM_{\rm baryon}}}} = (GM_{\rm baryon}a_\varphi)^{1/4} \tag{S44}$$

The radial coordinate r cancels out identically, locking the outer galactic disk into a strictly flat rotation curve and analytically recovering the exact Baryonic Tully–Fisher Relation ($V_\infty^4 = GM_{\rm baryon}a_0\varphi$) [15, 16].

(b) High-Acceleration Solar System Limit ($a_{\rm baryonic} \gg a_\varphi$) — Compliance with Cassini PPN Bounds: Within the inner Solar System at $r = 1$ AU $= 1.496\times10^{11}$ m, the solar Newtonian gravitational acceleration is $a_\odot(1\text{ AU}) = \frac{GM_\odot}{r^2} \approx 5.93\times10^{-3}$ m s^{-2}, giving $x(1\text{ AU}) = (5.93\times10^{-3}/1.8264\times10^{-10})^{1/2} \approx 5.698\times10^3$. The fractional deviation of the Phantom Metric from standard Einsteinian General Relativity ($\Delta\nu \equiv \nu - 1$) is suppressed exponentially:

$$\Delta\nu(1\text{ AU}) = \frac{\exp(-x)}{1-\exp(-x)} \approx \exp(-5698) \approx 10^{-2474} \tag{S45}$$

Even in the outer Solar System at the orbit of Neptune ($r = 30$ AU, where $a_{\rm baryonic} \approx 6.59\times10^{-6}$ m s^{-2}), we find $x(30\text{ AU}) \approx 190.0$, yielding a Parameterized Post-Newtonian (PPN) deviation of $|\Delta\gamma_{\rm PPN}| \sim \exp(-190) \approx 3.0\times10^{-83} \ll |\gamma_{\rm PPN}-1|_{\rm Cassini} = 2.3\times10^{-5}$ [17]. Unlike algebraic MOND interpolation functions (where $\Delta\nu \sim a_0/a \sim 10^{-4}$ at Neptune), the exponential DBI saturation decay restores exact General Relativity throughout the Solar System by more than 78 orders of magnitude beyond Cassini spacecraft bounds.

Section S4: Null Geodesic Equation, Strong Lensing Aperture Derivation (R_E and R_{BB}), and Fermat's Principle for Quadrupoles and Broken Arcs

S4.1 Derivation of the Macroscopic Saturation Boundary (R_{BB}) and Einstein Ring Radius (R_E)

Consider an external relativistic photon ($ds^2 = 0$) traversing the gravitational field of a thermodynamically relaxed single-node system of total baryonic mass $M_{\rm baryon}$. Because the incident photon approaches from cosmological distances outside the internal disk plane, its optical interaction projects onto an absolute two-dimensional sky-plane aperture (geometric factor 1.0, independent of the internal aspect ratio η of the lens). The radial gravitational gradient of the baryonic lens attenuates as $a(r) = \frac{GM_{\rm baryon}}{r^2}$ until it intersects the fundamental cosmological horizon floor a_0:

$$a(R_{BB}) = \frac{GM_{\rm baryon}}{R_{BB}^2} = a_0 \implies R_{BB} = \sqrt{\frac{GM_{\rm baryon}}{a_0}} = \sqrt{\frac{2\pi GM_{\rm baryon}}{cH_0}} \tag{S46}$$

In the perturbed metric $ds^2 = a^2[-(1+2\Psi)d\tau^2 + (1-2\Phi)d\vec{x}^2]$, the null geodesic equation $k^\mu\nabla_\mu k^\nu = 0$ governing the transverse deflection of a light ray with impact parameter $\vec{\xi}$ in the lens plane is:

$$\frac{d^2\vec{\xi}}{dl^2} = -\frac{1}{c^2}\nabla_\perp(\Psi+\Phi) = -\nabla_\perp \ln n_{\rm eff}(\vec{\xi}, l) \tag{S47}$$

where $n_{\rm eff} = 1 - \frac{\Psi+\Phi}{c^2}$ is the effective optical refractive index of spacetime. While classical General Relativity integrates weak-field potentials assuming an unperturbed Euclidean background, the DBI saturation factor $\gamma = (1-2|X|/M^4)^{-1/2}$ undergoes a phase transition along the two-dimensional boundary $r_\perp = |\vec{\xi}| = R_{BB}$. Integrating the resulting logarithmic saturation potential $\Phi_{\rm sat}(r) = V_\infty^2 \ln(r/R_{BB})$ (with $V_\infty^4 = GM_{\rm baryon}a_0$) along the line of sight via Eq. (S47) yields the parameter-free relativistic deflection angle $\hat{\alpha} = \frac{4\pi V_\infty^2}{c^2} = \frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2} = \frac{4\pi a_0 R_{BB}}{c^2}$ and the physical lens-plane Einstein ring radius:

$$R_{E,\rm calc} \equiv D_L\theta_{E,\rm calc} = D_L\frac{D_{LS}}{D_S}\frac{4\pi\sqrt{GM_{\rm baryon}a_0}}{c^2} \tag{S48}$$

where D_L, D_S, D_{LS} are the angular diameter distances and θ_E is the observed angular Einstein radius in radians [18–22, 27].

Optical Waveguide Capture Condition and Empirical Verification (SLACS and CLASH): Physically, the refractive transition at $r_\perp = R_{BB}$ traps incident rays into an Einstein ring whenever the background source lies within the cosmological waveguide capture cone ($D_{LS}/D_S \gtrsim 0.5$, satisfied across strong-lens surveys; whereas for $D_{LS}/D_S \to 0$, no ring forms). Evaluated across all $N = 110$ confirmed Grade-A galaxy-scale strong lenses in the combined SLACS IX ($N = 70$) [20] and SLACS XIII S4TM ($N = 40$) [21] catalogs with zero free parameters (0% dark matter), Eq. (S48) reproduces the HST observed physical Einstein radii ($R_{E,\rm obs} = D_L\theta_E$) with $R^2 = 0.8261$ ($p \approx 1.6\times10^{-27}$, mean ratio 1.042)

across the homogeneous multi-band SLACS IX subset and $R^2 = 0.7955$ ($p \approx 5.1 \times 10^{-39}$, strict 1:1 parity $R^2_{1:1} = 0.7773$, mean ratio 1.016 ± 0.198, median 1.028, and 98.6% ensemble sum parity: 370.3 kpc observed vs. 375.7 kpc predicted) across the full $N = 110$ sample, while at the cluster scale (CLASH, where $R_{E,\mathrm{calc}}$ coincides directly with R_{BB} at the cluster baryon fraction $f_b \approx 13\%$) it achieves $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$). Because in massive elliptical galaxies R_{BB} lies close to the baryonic effective radius ($R_{\mathrm{eff}} \approx R_E$), this phase transition naturally accounts for the apparent departure from a pure 3D power-law profile ($\lambda_{\mathrm{int}} \neq 1$) observed in joint JWST/Keck lensing-kinematic analyses [27], while beyond R_{BB} the continuous radial attenuation $\nu(r) = \{1 - \exp[-(a_{\mathrm{baryonic}}(r)/a_0)^{1/2}]\}^{-1}$ governs extended cluster weak-lensing shear profiles out to megaparsec scales [30].

S4.2 Analytical Proof of the Einstein Cross and Fragmented Optical Arcs via Fermat's Principle

By Fermat's principle in General Relativity, observed lensed images form at angular coordinates where the optical arrival-time functional $T(\vec{\theta})$ is stationary ($\nabla_{\vec{\theta}} T(\vec{\theta}) = 0$). Along the saturated boundary perimeter $r_\perp = R_{BB}$, the azimuthal variation of the arrival-time functional for a source aligned with the optical axis is governed by the multipole expansion of the projected baryonic potential $\Phi_{\mathrm{eff}}(R_{BB}, \theta)$ over the azimuthal angle $\theta \in [0, 2\pi)$:

$$\Phi_{\mathrm{eff}}(R_{BB}, \theta) = \Phi_0(R_{BB}) \Bigg[1 + \varepsilon_2 \cos(2\theta) + \sum_{m=3}^{\infty} \varepsilon_m \cos(m(\theta - \theta_m)) \Bigg] \tag{S49}$$

where ε_2 represents the dimensionless baryonic quadrupole ellipticity and ε_m ($m \geq 3$) represent higher-order morphological asymmetries. Imposing Fermat's stationarity condition along the saturation perimeter ($\frac{\partial T}{\partial \theta} \propto -\frac{\partial \Phi_{\mathrm{eff}}}{\partial \theta} = 0$) yields:

$$\frac{\partial \Phi_{\mathrm{eff}}}{\partial \theta} = -\Phi_0(R_{BB}) \Bigg[2\varepsilon_2 \sin(2\theta) + \sum_{m=3}^{\infty} m \varepsilon_m \sin(m(\theta - \theta_m)) \Bigg] = 0 \tag{S50}$$

This single equation analytically generates all three observed strong-lensing topologies without dark matter:

1. **Axisymmetric Baryonic Lens ($\varepsilon_2 = 0, \varepsilon_m = 0$):** The azimuthal derivative vanishes identically ($\frac{\partial \Phi_{\mathrm{eff}}}{\partial \theta} \equiv 0$) for all $\theta \in [0, 2\pi)$, focusing light into a complete, continuous Einstein Ring of radius R_E governed by R_{BB}.

2. **Symmetric Elliptical / Quadrupolar Lens ($\varepsilon_2 \neq 0, \varepsilon_{m \geq 3} = 0$):** The stationarity condition reduces to $-2\varepsilon_2 \Phi_0 \sin(2\theta) = 0 \implies 2\theta = k\pi$ ($k = 0, 1, 2, 3$), yielding four discrete stationary optical nodes separated by $\pi/2$ (90°): $\theta_1 = 0, \theta_2 = \frac{\pi}{2}, \theta_3 = \pi, \theta_4 = \frac{3\pi}{2}$. Evaluating the second derivative $\frac{\partial^2 \Phi_{\mathrm{eff}}}{\partial \theta^2} = -4\varepsilon_2 \Phi_0 \cos(2\theta)$ confirms that $\theta = 0, \pi$ are arrival-time minima and $\theta = \pi/2, 3\pi/2$ are saddle points, analytically reproducing the quadruple image geometry of the Einstein Cross [22].

3. **Asymmetric / Unrelaxed Baryonic Lens ($\varepsilon_{m \geq 3} \neq 0$):** Localized higher-order multipoles in the visible baryonic mass distribution—including compact, dust-obscured giant molecular clouds (GMCs, $\sim 10^5$–$10^7\, M_\odot$ within 10–140 pc), star-forming clumps, and wandering black holes [28, 29]—fragment the stationary locus into discrete angular intervals satisfying $\frac{\partial \Phi_{\mathrm{eff}}}{\partial \theta} \approx 0$, deterministically breaking the R_{BB} perimeter into fragmented optical arcs and localized millilensing perturbations without invoking invisible NFW dark matter subhalos.

Section S5: Statistical Audit Formulation and Primordial Cosmological Continuity (ΛTSM)

S5.1 SI Unit Conversion, Dimensional Homogeneity, and Zero-Parameter 1:1 Goodness-of-Fit

1. Exact SI Kinematic Conversion (SPARC Audit): For a total effective baryonic velocity $V_{\mathrm{baryonic}}(r) = (V^2_{\mathrm{gas}} + \Upsilon_* V^2_{\mathrm{disk}} + \Upsilon_* V^2_{\mathrm{bulge}})^{1/2}$ given in $\mathrm{km\,s^{-1}}$ ($1\ \mathrm{km\,s^{-1}} = 10^3\ \mathrm{m\,s^{-1}}$) at galactocentric radius r given in kpc ($1\ \mathrm{kpc} = 3.08567758 \times 10^{19}$ m), the exact Newtonian baryonic acceleration in SI units ($\mathrm{m\,s^{-2}}$) is [23]:

$$a_{\mathrm{baryonic}}(r) = \frac{(10^3\, V_{\mathrm{baryonic}}(r))^2}{3.08567758 \times 10^{19}\, r} \tag{S51}$$

2. Strict 1:1 Parity Coefficient of Determination ($R^2_{1:1}$) and p-value Formulation: Unlike standard linear regression, which optimizes two free parameters (slope β_1 and intercept β_0), our zero-parameter audit evaluates model predictions $y_{\mathrm{calc},i}$ directly against the strict identity line $y_{\mathrm{obs},i} = y_{\mathrm{calc},i}$ ($y = x$):

$$R^2_{1:1} \equiv 1 - \frac{SS_{\mathrm{res}}}{SS_{\mathrm{tot}}} = 1 - \frac{\sum_{i=1}^{N} (y_{\mathrm{obs},i} - y_{\mathrm{calc},i})^2}{\sum_{i=1}^{N} (y_{\mathrm{obs},i} - \bar{y}_{\mathrm{obs}})^2} \tag{S52}$$

where $\bar{y}_{\mathrm{obs}} = \frac{1}{N} \sum_{i=1}^{N} y_{\mathrm{obs},i}$. The two-tailed statistical significance (p-value) for sample size N with $\nu_{\mathrm{df}} = N - 2$ degrees of freedom is computed from the exact Student's t-distribution integral:

$$t_{\mathrm{stat}} = r \sqrt{\frac{N-2}{1-r^2}}, \qquad p = 2 \int_{|t_{\mathrm{stat}}|}^{\infty} \frac{\Gamma\left(\frac{\nu_{\mathrm{df}}+1}{2}\right)}{\sqrt{\nu_{\mathrm{df}} \pi}\, \Gamma\left(\frac{\nu_{\mathrm{df}}}{2}\right)} \left(1 + \frac{x^2}{\nu_{\mathrm{df}}}\right)^{-\frac{\nu_{\mathrm{df}}+1}{2}} dx \tag{S53}$$

Across all $N = 3,391$ discrete SPARC radial measurements ($\nu_{\mathrm{df}} = 3,389$), Eq. (S53) yields $p < 10^{-300}$ for both the locked ($\Upsilon_* = 0.46, R^2 = 0.9150$) and stellar-synthesis relaxed ($R^2 = 0.9698$) audits. Even under the ultra-conservative convention of treating each of the $N = 175$ galaxies as a single independent degree of freedom ($\nu_{\mathrm{df}} = 173$, removing all intra-galaxy radial autocorrelation), Eq. (S53) yields $t_{\mathrm{stat}} = 43.15$ ($p \approx 1.1 \times 10^{-95}$) for the locked baseline and $t_{\mathrm{stat}} = 74.53$ ($p \approx 4.8 \times 10^{-132}$) for the relaxed audit. Likewise, across the $N = 110$ confirmed Grade-A strong-lensing galaxies in the combined SLACS IX ($N = 70$) [20] and SLACS XIII S4TM ($N = 40$) [21] catalogs, Eqs. (S52)–(S53) yield $R^2 = 0.7955$ ($t_{\mathrm{stat}} = 20.50, p \approx 5.1 \times 10^{-39}$, with strict 1:1 parity $R^2_{1:1} = 0.7773$ and $R^2 = 0.8261, p \approx 1.6 \times 10^{-27}$ for the $N = 70$ multi-band SLACS IX subset), while the morphologically relaxed CLASH galaxy clusters achieve $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$).

S5.2 Early-Universe Limit: Covariant Unification with Primordial Cosmology (ΛTSM) and the CMB Acoustic Power Spectra

To demonstrate rigorous theoretical continuity between late-universe galactic equilibrium ($z \approx 0$) and early-universe primordial cosmology ($z \approx 1100$), we evaluate the four-dimensional DBI action and its second-order perturbed Lagrangian (Section S3.1) in a spatially homogeneous FLRW background $ds^2 = a^2(\tau)[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j]$ across comoving Fourier wavenumber k [24–26].

1. Background Topological Density Ratio ($\omega_{\text{phantom}}/\omega_b$): In the homogeneous cosmological background ($2\bar{X} \to M^4, \bar{\gamma} \to \infty$), the effective equation of state vanishes ($w_{\text{eff}} = P_\phi/\rho_\phi = \bar{\gamma}^{-1} \to 0$) and the adiabatic sound speed vanishes ($c_s^2 = \bar{\gamma}^{-2} \to 0$), so the background topological energy density scales identically to pressureless comoving matter ($\rho_{\text{phantom}} \propto a^{-3}$, with synchronous density contrast $\delta'_{\text{phantom}} = -\frac{1}{2}h'$ [13]). Equipartition of the $D^2 = 4^2 = 16$ algebraically independent directional flux components of the four-dimensional orthonormal tetrad frame field (vierbein) e^a_μ on the tangent bundle $T\mathcal{M}_4$ relative to the $d = D-1 = 3$ spatial hypersurface degrees of freedom ($\frac{16}{3}$), combined with the one-loop four-dimensional hyperspherical horizon impedance correction ($(4\pi)^{-D/2} = \frac{1}{16\pi^2}$), fixes the exact parameter-free background density ratio:

$$\frac{\omega_{\text{phantom}}}{\omega_b} = \frac{16}{3}\left(1 + \frac{1}{16\pi^2}\right) \approx 5.367107 \tag{S54}$$

For the Planck 2018 baryonic density $\omega_b = 0.02237$ [24], this yields $\omega_{\text{phantom}} = 0.120062$, matching the empirical ΛCDM dark matter density ($\omega_{\text{cdm}} = 0.1200 \pm 0.0012$) to within 0.05% with zero free parameters.

2. Decoupling Acoustic Phase Transition (`tca_off`): During the deep radiation-dominated tight-coupling epoch ($a \ll a_{\text{eq}}$, `tca_on`), photons and baryons are locked into a single relativistic fluid ($R \gg 1, \theta_b \approx \theta_\gamma$), preventing independent baryonic slip. As the plasma enters the decoupling transition (`tca_off`, $a \sim a_{\text{eq}} \to \pi a_{\text{eq}}$), longitudinal baryonic compression modes couple to the trace-free 3D Maclaurin spatial shear operator ($\mathcal{K}_0 = 1 - \frac{1}{3} = \frac{2}{3}$) and the causal horizon surface impedance ($\Gamma_0 = \frac{1}{8\pi}$). Enforcing strict super-horizon causality via the Helmholtz–Yukawa projection operator $\Pi_{\text{causal}}(k,\mathcal{H}) = \frac{k^2}{k^2+\mathcal{H}^2}$ and post-recombination neutralization freeze-out via $\Theta_{\text{rec}}(a) = \exp\left[-\left(\frac{a}{\pi a_{\text{eq}}}\right)^4\right]$ yields the parameter-free baryonic Euler momentum equation:

$$\theta_b' = -\left[\mathcal{H} + \Gamma_{\text{active}}(k,a,\theta_b)\right]\theta_b + \left[c_s^2 k^2 - \mathcal{K}_{\text{phantom}}(k,a)\right]\delta_b + k^2\Psi + R\tau_c^{-1}(\theta_\gamma - \theta_b) \tag{S55}$$

where:

$$\mathcal{K}_{\text{phantom}}(k,a) = \frac{2}{3}\mathcal{H}^2\left(\frac{k^2}{k^2+\mathcal{H}^2}\right)\Theta_{\text{rec}}(a) \tag{S56}$$

$$\Gamma_{\text{active}}(k,a,\theta_b) = \frac{1}{8\pi}\left(\frac{a_{\text{eq}}}{a}\right)\left(\frac{k^2}{k^2+\mathcal{H}^2}\right)\Theta_{\text{rec}}(a)\,\mathcal{S}(\theta_b) \tag{S57}$$

with $\mathcal{S}(\theta_b) \equiv \frac{1}{2}[1 + \tanh(\theta_b/\theta_{\text{tol}})]$ ($\theta_{\text{tol}} = 10^{-12}$ Mpc^{-1}) denoting the smooth C^∞ covariant expansion gate.

- **In the Decoupling Primordial Regime ($a_{\text{eq}} \lesssim a \lesssim \pi a_{\text{eq}}, z \approx 3400 \to 1100$):** The elastic curvature term $\mathcal{K}_{\text{phantom}}$ stabilizes the acoustic gravitational potential wells, while the directional horizon braking Γ_{active} regulates rarefaction peaks, reproducing the Planck 2018 CMB TT, TE, EE power spectra ($R^2_{TT} = 0.9970, R^2_{TE} = 0.9951, R^2_{EE} = 0.9951$) without cold dark matter.

- **In the Post-Recombination and Late-Universe Regime ($a \gg \pi a_{\text{eq}}, z \to 0$):** The plasma phase-transition gate $\Theta_{\text{rec}}(a)$ freezes out acoustic damping, preserving the exact linear matter clustering amplitude ($\sigma_8 = 0.8229$) on large scales while reducing continuously to the quasi-static spatial saturation equations (Sections S1–S4) within virialized galactic and cluster structures below $a_0 = \frac{cH_0}{2\pi}$.

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